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Identifying Key Predictors of Cognitive Status Using Machine Learning Models in the Mexican Health and Aging Study

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Introduction

1.1 Key Words

Dementia, MHAS, Machine Learning, Random Forest,Predictor Identification

1.2 Motivation

According to Alzheimer's Disease International (ADI)[1, 2, 3], today there are more than 55 million people living with dementia as of 2024, and that number is expected to rise to 139 million by 2050. Dementia, on it's own, is a term used to describe a variety neurodegenerative diseases that primarily affect memory, thinking, behavior, and emotion. Dementia manifests itself with a wide array of symptoms, such as memory loss, language problems, disorientation, etc. In the words of [4]. ≪The ability to predict dementia incidence is critically relevant for decisions of clinicians to manage patients in follow-up, and for investigators to recruit participants into clinical trials. Early precise prevention and intervention targeting the highly suspected population can effectively reduce the disease burden and save enormous medical resources from those unlikely to progress to dementia≫. In order to do so, several Risk Prediction Models have been developed over the years, such as the Cardiovascular Risk Factors, Aging, and Incidence of Dementia (CAIDE)[5], the Australian National University Alzheimer's Disease Risk Index (ANU-ADRI) [6], or the recently developed UKBioBank dementia risk prediction (UKB-DRP)[4]. In order to create a new prediction model, a data set of the right quality and the correct quantity must be found to train and test it.

1.3 Problem description

The UKB-DRP was previously mentioned as one of the Risk Prediction Models developed in order aid on such task. It's novelty lies on the methods with which it was created, Machine Learning (ML) [4]. They implemented an open source ML framework called LightBGM with the task to classify the participants into several groups depending on the risk factor they might fall in. However, the problem with ML lies on the amount and quality of the data that it is fed in order to generate an accurate prediction. There is the intention to develop a new model based on the data stored on the Mexican Health and Aging Study (MHAS) data base [7], if that model also uses ML, or any other type of algorithm that works similarly (such as genetic algorithms) on it's implementation, it would also rely heavily on it. But how to ensure the quality of the data sets that will be fed to the algorithm? Also, considering the amount of data that lies stored on a Longitudinal Data Base and the amount of said DB's to be used, how does one search for trending variables or any other indicator that the data will be of quality? That is where our research comes in.

1.4 Research question

How can we ensure the quality of a training data set based on the Mexican Health and Aging Study's [7] Database?

1.5 Hypothesis

According to IBM [8], Machine Learning is one of the best tools when it comes to data analysis, particularly shining on big datasets such as MHAS'. However, "Machine Learning" is a broad term in on itself, as it can contain several algorithms for data processing. On this study, we have used a type of Random Forest, Historic Random Forest, along with a library called Shapley Additive exPlanations, or SHAP.

1.6 General objectives

To identify the most important predictors for dementia prediction using a Machine Learning Algorithm by analyzing their relevance through time, thus ensuring the data's quality.

1.7 Particular objectives

The more specific steps required to accomplish the General objective are the following:

- Identify the top 5 most important predictors across the different waves of application
- Follow the evolution and change of this importance throughout the 4 waves of application

1.8 Contributions

By identifying the most important variables we can ensure that any future predictive models that aim to be trained on the MHAS's [7] DB, we can make their development process more efficient and their prediction more accurate.

Chapter 2

Background

2.1 Longitudinal Databases and MHAS

The ENASEM's Data Base [7], along with the other DB's that will be analyzed during this research must be of a certain type: longitudinal. According to [9], when speaking on a statistic sense, "A longitudinal study is one that involves more than two measurements throughout a follow-up period; there must be more than two, since every cohort study has this number of measurements, one at the beginning and one at the end of the follow-up period." These type of studies imply repeated measurements on the participants across time, which means they must be managed with a close relation with time.

MHAS, [7], is a very good example to develop such concept, as it is a national longitudinal study that includes people over 50 years old. It's first wave took place on 2001 and it's initial population included populations from both rural and urban areas. The survey was followed up in the years 2003, 2012, 2015, 2018, 2021 and 2024, with 2012 and 2018 being the years in which new populations of adults were added, this time born in the periods 1952-1962 and 1963-168 respectively. This study is the product of the efforts of researchers belonging to the University of Texas' Medical Branch (UTMB), the Instituto Nacional de Estadística y Geografía (INEGI, México), the Instituto Nacional de Geriatría (INGER, México), the Instituto Nacional de Salud Pública (INSP, México), the University of Columbia and the University of California Los Angeles (UCLA).

2.2 Predictive Models

We have also talked about several Predictive Models, such as the UKB-DRP [4], which used a Machine Learning Algorithm on the UK BioBank's database, which has around 502,414 participants and more than ten years of follow-up. As of now, it only works appropriately with Alzheimer's Disease, but it's authors mention the desire to update the model for all types of dementia. We have mentioned also the Australian National University Alzheimer's Disease Risk Index (ANU-ADRI), [6]. This model includes 15 components: age, sex, education, body mass index (BMI), depression, diabetes, cholesterol, smoking, traumatic brain injury, physical activity, cognitive activity, social engagement, alcohol intake, dietary fish intake, and pesticide exposure. in order to make it's prediction. It does this by allocating specific points, which are weighted relative to the factor's effect size, based on self-reported data. The ANU-ADRI's components were selected by following this process: First, an initial list of potential risk factors for Alzheimer's Disease (AD) and cognitive decline was compiled. Afterwards, risk and protective factors with strong evidence of association with Alzheimer's Disease (AD) were selected from the initial list. Inclusion required support from high-quality meta-analyses or multiple cohort studies with pooled effect sizes. Next odd ratios for risk and protective factors were sourced using a hierarchical approach. Priority was given to systematic reviews with clearly defined exposures. Finally, exposure variable definitions were established for inclusion in the ANU Alzheimer's Disease Risk Index, guiding the selection of questionnaire items. Said features can be seen more clearly on the following table:

Table 2.1: Comparison between the UKB-DRP and ANU-ADRI Predictive Models for Alzheimer's Disease Risk

Feature	**UKB-DRP (UK BioBank Dementia Risk Prediction)**	**ANU-ADRI (Australian National University Alzheimer's Disease Risk Index)**
Source / Database	UK BioBank	Post hoc analysis based on the FINGER trial (Finland, 2009–2011)
Sample Size	$\sim$502,414 participants	1174 participants
Follow-up Duration	More than 10 years	2 years (intervention period)
Main Focus	Alzheimer's Disease (currently; plans to expand to all dementia types)	Assessment of Alzheimer's Disease risk and cognitive decline
Methodology	Machine Learning algorithm	Point-based scoring system using self-reported and clinical data
Model Inputs / Components	Ten predictors identified from UK BioBank data (biomedical, demographic, and lifestyle features); missing values handled natively by LightGBM	15 components: age, sex, education, BMI, depression, diabetes, cholesterol, smoking, traumatic brain injury, physical activity, cognitive activity, social engagement, alcohol intake, dietary fish intake, and pesticide exposure
Data Type	Large-scale biomedical and clinical data	Self-reported and clinical baseline data collected from FINGER participants
Weighting / Scoring	Determined through machine learning feature weighting	Each factor assigned a weighted point value relative to its effect size, based on established epidemiological evidence
Model Development Process	Built using UK BioBank dataset with ML techniques	developed through systematic literature review and meta-analyses
Validation / Evidence Base	Based on UK BioBank follow-up data	Validated within the FINGER randomized controlled trial
Future Plans / Adaptability	To be expanded to predict all types of dementia	Demonstrated applicability in multidomain interventions; supports integration into preventive lifestyle and cognitive training programs

2.3 A new Model: Selected Variables

There is the intention of developing a new Prediction model based on the MHAS database. From it, some variables have been selected in order to feed them to said model based off of Lancet's Risk Factors [10]. Said variables are:

- Education
- Hearing loss
- Cholesterol
- Brain injury
- Physical activity
- Diabetes
- Smoking
- Hypertension
- Obesity
- Alcohol consumption
- Social isolation
- Air pollution
- Vision loss

However, for this research, some other risk factors have been selected based on several other studies, like Guo et al.'s [11], Zhang et al's. [12], Halse et al.'s [13] and Luck et al.'s[14]. Thanks to these studies, the following risk predictors were selected:

- Stress
- Locus of control
- Time use
- Falls
- Hospitalization

2.4 Data Interpretation Tools: Machine Learning

In order to choose wisely our final tool prospect(s), we need to properly define the broader concept of "what is Machine Learning". According to IBM and Tantawi [8, 15], Machine Learning, or ML for short, is a branch of Artificial Intelligence focused on emulating the way a human brain learns on a computer. Normally, ML algorithms include these 3 steps: a decision process on which a prediction or classification is made, an error function that evaluates said decision process and an optimization function where the model better readjusts itself to fit the data points if necessary. We have called it a broader concept, as there are more than one algorithm that can be used on ML; a good example are Artificial Neural Networks, which mimic the function of the biological brain, being designed to model the relation among the input and output parameters through a training process. ML is great for data analysis, being particularly great at finding trends on massive volumes of it. It has one major downside, however, as it's correct operation requires equally numerous data sets that must be precise and impartial. For this research, the ones that were more heavily looked into where Random Forests, Linear Regression and Logistic Regression.

According to GeeksforGeeks [16], Random Forest, or RF, is an algorithm that is based off decision trees. This is an algorithm that generates several of these that analyze random parts of the data set. Once these are finished, the algorithm votes on the most optimal path to follow. By combining data from each tree, the risk of overfitting is significantly reduced. Also, the Random Forest method excels when it comes to handling missing data. It shows which columns are most important for prediction, is good at handling large and complex data, and finally, combining the trees improves the quality of the prediction.

Next up, Linear Regression. Based off of GeeksforGeeks' definition [17], it is a type of supervised machine learning algorithm. This type of regression learns from labeled data sets and assigns the data points with the most optimized linear functions, allowing predictions to be made on new data sets. One of its main disadvantages is that it assumes a linear relationship between input and output, meaning that the output changes at a constant rate as the input changes, making it very prone to overfitting.

Finally, Logistic Regression, as GeeksforGeeks puts it [18], is more commonly used for classification problems. Instead of using a linear function, it uses a sigmoid (s-shaped) function. Because of this, it's not as susceptible to many of the problems that linear regression is. It's excellent at identifying anomalous data, such as fraudulent activity in the banking sector. Although it may be a bit closer to what we're looking for in this project, logistic regression can fall a bit short when the data set is too large.

To better visualize the differences between these 3 algorithms, here's a table:

Table 2.2: Comparison of Machine Learning Algorithms: Random Forest, Linear Regression, and Logistic Regression

Feature	**Random Forest (RF)**	**Linear Regression (LR)**	**Logistic Regression (LogR)**
Type of Algorithm	Ensemble method based on Decision Trees	Supervised machine learning algorithm (regression)	Supervised machine learning algorithm (classification)
Core Mechanism	Builds multiple random decision trees on subsets of the data and aggregates their predictions by voting to reduce overfitting	Fits a best-fit straight line to minimize the difference between actual and predicted values	Uses a sigmoid (S-shaped) function to estimate probabilities and classify outcomes
Strengths	Handles missing data well; reduces overfitting; identifies feature importance; robust with large and complex datasets	Simple to implement and interpret; effective for linearly related variables; useful in forecasting and trend analysis	Effective for classification problems; resistant to overfitting; good at detecting anomalies (e.g., fraud detection)
Weaknesses	Can be computationally intensive; less interpretable due to multiple trees	Assumes linearity between input and output; sensitive to outliers; prone to overfitting if not regularized	Performance decreases with very large datasets; assumes linear decision boundary in transformed (log-odds) space
Common Applications	Feature selection, ensemble prediction, risk modeling, and biomedical data analysis	Marketing, finance, and trend forecasting	Fraud detection, medical diagnosis, and binary classification tasks

Chapter 3

State of the art

3.1 UKB-DRP

As was previously stated, ML has already seen its use on prediction models that are centered on dementia, [4]. The UKB-DRP took over 500,000 people aged between 40 and 69 for it's initial evaluation, which took place between the 1st of March of 2006 and the 31st of October of 2010. The study consisted of interviews, questionnaires, and other lifestyle and health assessments, physical measurements, biological sample analysis, imaging, and genotyping. Those who already had dementia or stroke, or who had no follow-up records, were excluded. The total number of participants was 425,159. First, only the medically relevant variables were included where a preliminary screening phase took place to exclude non-informative variables whose missing values exceeded 40 percent of all participants, and procedural metric variables that lacked clinical relevance were manually refined; thus there remained 366 indicators left which included data such as demographic characteristics, recorded lifestyle and health information, cognitive function tests, and biological sample analysis. Data was extracted from reported first occurrences of dementia, algorithmically defined, documented from death registers, abstracted from inpatient data, and recorded from primary care settings.

This is where ML comes in. The final predictors were selected this way: first, a naive classifier called LightGBM6 was established, and each feature was ranked according to its contribution to model performance based on information gain, which can be regarded as the predictor's ability to identify future incidence of dementia. From this classification, 50 indicators were selected and ordered according to Spearman's range. Then, a progressive sequential selection strategy was employed, in which features from the preselected subset were reclassified according to a newly developed classifier. Finally, consecutive classifiers were developed with sequentially added predictors based on the importance ranking orders of the updated predictors. In total, 10 predictors were identified for model development.

3.2 Uses of Random Forest

Qin et al.'s [19] study used a Random Forest algorithm as part of an integrated bioinformatics pipeline to identify and prioritize potential protein biomarkers involved in renal carcinoma. After performing transcriptomic differential expression and constructing a protein–protein interaction (PPI) network, the researchers used Random Forest models trained on hub gene expression profiles to evaluate their predictive relevance in distinguishing tumor versus normal tissue. The algorithm quantified gene importance through the Mean Decrease Gini index, revealing that ASPM, TOP2A, and KIF4A were the top predictors, explaining 84% of model variance. These genes were found to be strongly co-expressed, linked to mitotic regulation and epithelial–mesenchymal transition, and associated with poorer overall survival, confirming their potential as diagnostic and prognostic biomarkers in renal cancer.

In Nam and Song's [20] educational data science study, Random Forest was employed to predict students' persistence in online learning using large-scale data from 26,867 learners at a Korean open university. The model incorporated variables such as academic achievement, learning time, course completion rates, and number of logins to the online learning system. Implemented via Python's Scikit-learn with 10,000 trees, the Random Forest achieved 85.46% classification accuracy and an F1 score of 0.90, indicating high predictive reliability. To interpret model outputs, the authors used SHAP (SHapley Additive exPlanations) values, identifying that academic achievement and engagement metrics after the 7th week (e.g., logins and learning time) were the strongest predictors of persistence. The findings suggest that Random Forest models can help institutions identify at-risk students early and design targeted interventions to enhance retention and academic success

Aside from that, we found that RF works very well with longitudinal DB's thanks to Hu y Szymczak's research [21]. The study presents several options for such adaptation: for longitudinal data with univariate and multivariate responses. The study mentions using Historical RFs, which incorporate the history of each predictor (past values summarized with functions such as counts in time windows) along with the current value to capture trajectories and not just point values. Other options they propose are using Mixed Effects Random Forests (MERF) to create a hybrid model. The main idea of doing this is to extend classical mixed models, which already model within-subject correlation through random effects and covariance structures. This can be done by replacing the linear part of the fixed effects in a LMM/GLMM (Generalized Linear Mixed Models) with Random Forests.

3.3 Uses of Linear Regression

The team of El Jihaoui et al. [22] have used Linear Regression as part of a combined statistical approach with Factor Analysis for Mixed Data (FAMD) to investigate the determinants of middle school students' academic performance in Morocco. After identifying four underlying latent factors — prior academic performance, academic delay, socioeconomic status, and class environment — through FAMD, the researchers applied Multiple Linear Regression (MLR) to quantify how each factor influenced cumulative grade point average (CGPA). The model was based on an extensive dataset of over 1 million observations, incorporating both quantitative and qualitative educational variables. The regression analysis revealed a highly predictive model, explaining 88.53% of the variance in CGPA, with prior academic performance emerging as the strongest predictor (76.6% of the explained variance), followed by academic delay (14.34%), socioeconomic status (6.02%), and class environment (3.03%). All predictors were statistically significant (p ¡ 0.001), highlighting the model's robustness in identifying students at risk and informing targeted educational interventions.

Watada et al. [23] have used it too. They proposed an advanced extension of Linear Regression called Type-2 Fuzzy Linear Linguistic Regression (T2F-LLR) to model uncertainty and vagueness in real-world decision-making scenarios. The model integrates credibility theory with fuzzy set mathematics to perform regression on linguistic and uncertain data. Instead of relying on crisp numerical inputs, the authors used fuzzy linguistic questionnaires from consumers and experts in a cosmetic company's marketing planning case covering four product categories. The T2F-LLR algorithm transformed qualitative linguistic terms (e.g., "always," "frequently") into fuzzy variables and applied linear regression to estimate relationships under uncertainty. Using a heuristic computational approach, the model achieved a mean absolute percentage error (MAPE) of 7.97%, with all variables statistically significant.

Statistical tests such as one-way ANOVA (p = 0.15) and paired t-tests (p = 0.16) indicated no significant difference between observed and predicted values, confirming the reliability and interpretability of this fuzzy regression model for uncertain, language-based decision-making environments.

3.4 Uses of Logistic Regression

Yildiz and Buldur [24] used Binary Logistic Regression Analysis (LRA) to examine how pre-service teachers' demographic variables — specifically gender and grade level — influenced their agreement with ethical principles in educational assessment. The analysis was conducted using SPSS 26, following all statistical assumptions for LRA (sample size adequacy, independence, multicollinearity, and outliers). Separate logistic regression models were developed across different ethical scenarios, with the dependent variable representing teachers' agreement with expert opinions. The findings revealed that the models were statistically significant across multiple scenarios (e.g., χ^2(3) = 29.302, p ¡ .001 in Scenario 4; χ^2(3) = 41.168, p ¡ .001 in Scenario 8), and classification accuracy ranged between 63.8% and 76.9%. Gender-based analysis showed that males were consistently less likely than females to agree with expert ethical judgments, with odds ratios ranging from 0.519 to 0.715, indicating a 28–48% lower likelihood. In contrast, grade-level analysis showed that senior students (Grades 3–4) had higher odds of agreement in several scenarios, suggesting that ethical awareness tends to develop over time during teacher training.

On Yeşilaydın et al.'s [25] cross-national study, Logistic Regression Analysis was applied to determine which health resource variables most strongly influenced a country's efficiency in achieving high scores on the Healthcare Access and Quality (HAQ) Index. The dependent variable was a binary indicator of whether a country was "efficient" in healthcare access and quality, and independent variables included health expenditure per capita, health expenditure as a percentage of GDP, number of doctors, nurses/midwives, and hospital beds. The regression model was statistically significant (χ^2(5) = 27.299, p ¡ 0.05) and explained 19.7% of the variance in efficiency (Nagelkerke R^2 = 0.197). Among the predictors, health expenditure per capita and health expenditure as a percentage of GDP emerged as significant factors (p = 0.029 and p = 0.003, respectively), demonstrating that financial investment in healthcare was a key determinant of national performance in access and quality. The Hosmer–Lemeshow test (χ^2 = 2.467, p = 0.963) indicated a good model fit, supporting the reliability of the regression. The study concluded that efficient health financing mechanisms substantially improve healthcare accessibility and quality across income groups, with wealthier nations showing notably higher HAQ scores.

Chapter 4

Methods

4.1 Algorithm Selection

As mentioned before, the correct algorithm to use on ML makes quite the difference. Let's start with Linear Regression [17, 22, 23], although simple to implement, it best works for linearly related variables. This makes it very sensible to outliers and prone to overfitting. Logistic Regression [18, 24, 25] finds itself on a very similar situation; working wonders when talking about finding trends (as it works mainly on classification) and being particularly resistant to overfitting. However, it is not as well suited for large datasets as the other algorithms, which MHAS is. Finally, Random Forest [16, 19, 20] was determined to be the right choice. It works very well with large datasets thanks to the way it works, randomly analyzing parts of the dataset, as well as having good management of missing data. It must be mentioned that the one weakness RF has is that running it takes a lot of resources and time. Not only that, but as was mentioned, RF can work well with longitudinal data. For this research, we chose to go the Historic Random Forest [21] route for its easier implementation.

4.2 File Selection

MHAS has several data sets that could be used for this analysis, and these are spread out among the 6 waves the study has been applied on. The first dataset that was considered was the Raw files, which are made of the direct responses of the MHAS interview. Many variables on these files are codes pointing at the section and number the question was in, many of the actual variables we wanted to analyze being found across several questions. The time that would take to process the raw data into a more condensed version to use would take much time, and thus, the raw data files are discarded.

Fortunately, MHAS has already processed said data onto the files known as 'Constructed Data' [26]. These files have more condensed variables sorted into various sub-sections. However, while more usable, the Constructed Data files still present a lot of missing data (wether this might have been because the respondent denied to answer, it was a proxy interview, etc.). Random Forest can work well with this missing answers, but it would be more optimal to work with a more robust dataset.

This is what brings us to the 3rd set of files, the Harmonized Data [27]. These files were made for the GATEWAY TO GLOBAL AGING DATA (g2aging) project, meaning the data they contained is way-more processed. Some of the data present here has been imputed to stop having missing answers. MHAS provides the appropriate documentation CITE on how these imputations where made, which is why we know they used IVEware software, which implements a Sequential Regression Multiple Imputation (SRMI) procedure. Each variable with missing values is regressed on all other relevant variables in an iterative sequence until stable imputations are obtained. Imputations were performed separately by wave (2001, 2003, 2012, 2015, 2018, and 2021) and by interview type (direct or proxy). Both the treatment these data has and the way it's missing variables were managed make them the correct files to choose.

A thing we can't forget is that MHAS isn't a dementia study, it's an aging study. In on itself, the main data files lack the necessary variables to create a 'target variable' of dementia incidence. Another group of files MHAS provides are the Cognitive Function Measurements or CFM files [28]. Here, some variables more relevant to cognitive impairment and proper follow-up are registered, alongside out target variable: congitive_status_v01. It holds both cognitive status and cognitive status 01, but for comparison with other waves, the v01 one is recommended.

A final thing to take in note is the following: We have discarded the data present on wave 1 and 2 (2001 and 2003). The reasoning behind was that, in comparison to the later waves, these 2 have a lot of missing variables on both the Harmonized files [27] and the CFM files [28]

4.3 File Selection

4.3.1 Identifiers and Demographic Variables

Having our files chosen, the next step was to define the variables to extract and where would they fit. For the identification variables, and the bases of our data set, the following were chosen:

Variable	**Answer type**	**Description**
unhhidnp	Numeric	Unique Person Identifier.
rabyear	Numeric	Year of Birth.
rabmonth	Numeric	Month of Birth.
age_W	Numeric	The age of the respondent on wave W.
cognitive_status_W	Category	The cognitive status of the respondent on wave W.

Table 4.1: Identifier and Demographic Variables

The 3 age variables are so we can correlate our results with age (if possible) and better follow them throughout time.

4.3.2 Education

For the 'Education' risk factor, the following variables were chosen:

Variable	**Answer type**	**Description**
raedyrs	Numeric	Number of Years of Education.
raliterate	Category	If the Respondent knows how to read and write.
ranumerate	Category	If the Respondent knows how to count.

Table 4.2: Education Variables

These variables were chosen from the *Education* and *Literacy and Numbers Subsections* found on **Section A. Demographics, Identifiers, and Weights** from the MHAS Harmonized Datafiles [27]. readyrs was chosen for it is archived as a variable directly linked towards Education. Both raliterate and ranumerate were considered too due to their close relation towards education too, as the number of education years also depend on such abilities or they can serve as a proxy in case of no education years available.

4.3.3 Hearing Loss

For this risk Factor, the following variables were chosen:

Variable	**Answer type**	**Description**
rWhearing	Category	Self-rated hearing on W wave
rWhearaid	Category	If the Respondent wears hearing aid on W wave.

Table 4.3: Hearing Loss Variables

These 2 variables were chosen directly from the *Hearing* subsection from **Section B. Health** from the Harmonized Files [27].

4.3.4 Cholesterol

There only exists a single variable regarding Cholesterol itself, and it is found on the *Health Behaviors: Preventive Care* subsection on *Section B. Health*.

Variable	**Answer type**	**Description**
rWcholst	Category	Cholesterol on the last 2 years

Table 4.4: Cholesterol Variable

4.3.5 Brain Injury

For this Risk Factor, the following variables were selected:

Variable	**Answer type**	**Description**
rWstroke	Category	If the Respondent ever had a stroke
rWrxstrok	Category	If the Respondent takes medication for a stroke
rWrecstrok	Numeric	Age at the moment of the most recent stroke

Table 4.5: Brain Injury Variables

These variables were taken from **Section B. Health** from the Harmonized Files [27], however they are encountered on different sub-sections. rWstroke is found on the *Doctor Diagnosed Health Problems: Ever Have Condition* subsection, rWrxstrok in the *Doctor Diagnosed Diseases: Whether Receives Treatment or Medication for Disease* subsection, and rWrecstrok is on the *Doctor Diagnosed Diseases: Age of Diagnosis* subsection. These variables, although in different subsections, build up the whole picture regarding cardiovascular brain injury. Other Risk Factors will have the exact same condition, with their variables being spread across several subsections.

4.3.6 Physical Activity

This Risk Factor finds itself on a very similar situation as Cholesterol, with it only having 1 variable:

Variable	**Answer type**	**Description**
rWvigact	Category	If the Respondant engages in vigorous physical activity 3 or more times a week.

Table 4.6: Physical Activity Variable

This variable is extracted from the *Health Behaviors: Physical Activity or Exercise* subsection found in **Section B. Health**.

4.3.7 Diabetes

Diabetes is one of those risk factors whose variables are found on various subsections:

Variable	**Answer type**	**Description**
rWdiabe	Category	If the Respondent ever was diagnosed with diabetes
rWrxdiab	Category	If the Respondent takes medication for diabetes (oral or insulin)

Table 4.7: Diabetes Variables

rWdiabe can be found on the *Doctor Diagnosed Health Problems: Ever Have Condition* subsection and rWrxdiab on *Doctor Diagnosed Diseases: Whether Receives Treatment or Medication for Disease*. Both of these subsections are found on **Section B. Health**. Also on *Doctor Diagnosed Diseases: Whether Receives Treatment or Medication for Disease* a variable called rWrxdiabo is described. This variable states wether the participant takes or not oral medication for Diabetes. As this possibility is already covered on rWrxdiab, it was omitted.

4.3.8 Smoking

All of the variables pertinent to the construction of this Risk factor can be found on the *Health Behaviors: Smoking (Cigarettes)* subsection, on **Section B. Health**.

Variable	**Answer type**	**Description**
rWsmokev	Category	If the Respondent ever smoked
rWsmoken	Category	If the Respondent smokes now
rWsmokef	Numeric	Number of Cigarettes per day
rWsmokefm	Numeric	Number of Cigarettes per day when smoking most
rWstrtsmok	Numeric	Age when the Respondent started smoking
rWquitsmok	Numeric	Age when the Respondent stopped smoking

Table 4.8: Smoking Variables

These variables make the whole subsection, meaning none were deemed as ”unnecessary”.

4.3.9 Hypertension

These variables can also be found on *Doctor Diagnosed Health Problems: Ever Have Condition* and *Doctor Diagnosed Diseases: Whether Receives Treatment or Medication for Disease* on **Section B. Health**

Variable	**Answer type**	**Description**
rWhibpe	Category	If the Respondent ever had High Blood Pressure
rWrxhibp	Category	If the Respondent takes medication for High Blood Pressure

Table 4.9: Hypertension Variables

No more variables can be found on the Harmonized files [27] regarding this risk factor.

4.3.10 Obesity

The *BMI* subsection, found on **Section B. Health** has several variables, but for building the Risk Factor itself, only these were chosen:

Variable	**Answer type**	**Description**
rWbmi	Numeric	The BMI of the respondent
rWobese	Category	If the Respondent is obese

Table 4.10: Hypertension Variables

Only these 2 were chosen because the other variables contained on the subsection are more numeric and category variables that would help construct an answer for rWobese, such as rWweight, which describes the weight of the respondent on the current wave.

4.3.11 Alcohol

All of the variables pertinent to the construction of this Risk factor can be found on the *Health Behaviors: Drinking* subsection, on **Section B. Health**.

Variable	Answer type	Description
rWdrink	Category	If the Respondent drinks Alcohol
rWdrinkd	Numeric	Number of days a week the Respondent drinks
rWdrinkn	Numeric	Number of drinks per day
rWdrinkb	Category	If the Respondent ever binge drinks
rWbinged	Numeric	Number of days the Respondent binge drinks
rWcage	Numeric	Cage summary score

Table 4.11: Alcohol Variables

Although there are more variables described on this subsection, they suffer the exact same problem as the whole 1 and 2 waves. These variables, such as rWdrinkcr, are only available for waves 1.4, This means that, for our dataset, they would only be available for 2 waves. rWcage is one of these, yet it was chosen to remain for it was considered to be of aid when building the importance of the Alcohol Consumption risk factor. There exists a variable in the same state, but it appears on a later Risk Factor.

4.3.12 Social Isolation

This Risk Factor isn't found across the several subsections of the Harmonized Data File [27], which means it was constructed. These are the variables:

Variable	Answer type	Description
hWrural_m	Category	Size of the Respondent's locality of residence
rWrfcntpmx_m	Category	If the Respondent has frequent contact with friends and relatives by phone or email
rWcesd_m	Numeric	The sum of several depression indicators (such as if the Respondent feels happy, if they feel lonely, etc.)

Table 4.12: Social Isolation Variables

Although the variable hWrural has been already used before, hWrural_m itself doesn't describe the same thing. The first one states if the respondent lives on an urban or rural area, whereas the second describes the size of this locality instead of just stating if it's one or the other. These variables can be found on *Urban or Rural* on **Section A. Demographics, Identifiers, and Weights**. Next, rWrfcntpmx_m is found on the *Contact with Relatives and Friends by Phone, Email* subsection, in **Section G. Family Structure**. This variable directly relates to the contact the Respondent has with their family. On this same subsection some other variables are declared, but they describe weekly and monthly contact. As 'Frequent' can be a subjective term, it was decided to keep only that variable instead of taking both monthly and weekly contact. Finally, the rWcesd_m variable was taken from *Depressive Symptoms: CESD* in the **Section Q. Psychosocial** section. This last one works to tie together the other variables (size of locality and frequency of contact with family) with an emotional affection.

4.3.13 Air Pollution

There are no direct indicators that directly reference Air Pollution on the document. However, this Risk Factor was built directly upon two other variables that where already explained: hWrural and hWrural_m, described on Tables 4.1 and 4.12. The reason this was decided was due to the fact that urban areas are severely more polluted in comparison to rural areas.

4.3.14 Vision

All the variables used to build the Loss of Vision risk factor were directly taken from the *Vision* subsection in **Section B. Health**

Variable	Answer type	Description
rWsight	Category	Self-rated sight of the Respondent
Wglasses	Category	If the Respondent wears glasses

Table 4.13: Vision Variables

4.3.15 Stress

Although 'Stress' does appear as its own section (**Section M. Stress**), most of the variables it contains only are available for waves 2 and 3. The only variable available for the 6 waves is hWchdeathe, found on *Experienced Death of a Child*. Had this not been the case, many of these variables could've had worked for both Stress and Social Isolation. The rest of the variables were constructed this variables:

Variable	Answer type	Description
rWunemp	Category	If the Respondent is currently unemployed
rWjhoursd	Numeric	The hours the Respondent works at their main job
hWchdeathe	Category	If the Respondent has experienced the death of a child
rWcconc_m	Category	If the Respondent experienced stress due to COVID

Table 4.14: Stress Variables

The other variables chosen for stress are rWunemp, found on the subsection *Unemployment Status*, rWjhoursd found on the *Hours at Main Job* subsection, and both can be found in **Section H. Employment History**. Finally rWcconc_m corresponds ONLY to wave 6. It can be found on the *Stress Due to COVID* subsection, located in **Section R. COVID**. This is the variable mentioned previously. Although only present on the last wave, it was considered to be relevant as the COVID-19 pandemic may carry significant changes to the way the data behaves.

4.3.16 Locus of Control

This Risk Factor is yet another one that doesn't have a dedicated section, or subsection. However, the **Section Q. Psychosocial** does have several subsections dedicated to life satisfaction. These subsections describe variables such as if the Respondent feels they have 'ideal living conditions' or 'if they'd change almost nothing if they lived again'. It can be argued these variables take on a tone closely related to "if the Respondent feels they have control of their life", or at least related to their power to make decisions over their own life. These variables all boil down into a unique one, found in subsection *Single Life Satisfaction Question*. The other variable chosen is rWlsatsc3, which is the general life satisfaction score, found on *Satisfaction with Life Scale*.

Variable	Answer type	Description
rWsatlifez	Numeric	How much does the Respondent agree with the statement 'I am satisfied with my life'
rWlsatsc3	Numeric	Satisfaction with life score

Table 4.15: Locus of Control Variables

4.3.17 Time Use

Once again, this risk factor is one that isn't directly referenced on a section or subsection. However, it can be built once again. rWsocact_m can be found on subsection *Any Weekly Social Activities or Participate in Religious Groups* in **Section G. Family Structure**, and both rWwork and rWslfemp can be found in **Section H. Employment History**, in subsections *Currently Working for Pay* and *Whether Self-Employed* respectively.

Variable	Answer type	Description
rWsocact_m	Category	If the Respondent participates in frequent social activities
rWwork	Category	If the Respondent is currently working for pay
rWslfemp	Category	If the Respondent is self-employed

Table 4.16: Time Use Variables

These variables were chosen as they represent a fair amount of time. rWsocact_m is a variable that could be useful too for Social Isolation, but as it is described as "activities", it was determined it would be best placed here. The other two, rWwork and rWslfemp, represent if the Respondent uses their time working, and we considered wether the Respondent was self-employed or not to be of importance as it might represent an additional challenge for them.

4.3.18 Falls

All of these variables were taken from the corresponding subsection, *Falls* present in Section **B. Health**.

Variable	Answer type	Description
rWfall	Category	If the Respondent has fallen down in the past 2 years
rWfallnum	Numeric	Number of time sthe Respondent has fallen down on the past 2 years
rWfallinj	Category	If the Respondent was injured due to a fall

Table 4.17: Falls Variables

Although there are more variables described in this subsection, only these 3 were chosen. The remaining variables focus more on if the Respondent's hip was affected due to a fall.

4.3.19 Hospitalization

Once more, the variables were taken from the corresponding subsection, *Medical Care Utilization: Hospital*, in **Section C. Health Care Utilization and Insurance**.

Variable	Answer type	Description
rWhosp1y	Category	If the respondent has had a Hospital stay in the previous 12 months
rWhspnit1y	Numeric	How many nights has the respondent has stayed in the hospital in the previous 12 months

Table 4.18: Hospitalization Variables

There are several more subsections described on this section. These other subsections describe things as doctor visits, other medical care utilization, etc. As we only need wether the Respondent was hospitalized or not for this Risk Factor, the other variables were deemed unnecessary.

4.4 Data Processing

4.4.1 Pre-Processing

The first step in our analysis is to filter our data. There are some participants that we don't want, or we don't need. For example, we don't want any participant that doesn't have any follow-ups. The filters our participants must go through are the following:

- The respondent must be present on 2 or more waves
- The respondent must be 50 years old or older

Now, in order to have a more complete set of answers for each variables, the rows classified as 'Proxy Interview' were removed. This was a safe thing to do, as in no wave these types of interview reach over 8% of the answers:

- 2012: 6.6%
- 2015: 6.17%
- 2018: 7.78%
- 2021: 7.99%

After removing our proxy interviews, the 1st filter must still be valid in order to remain.

Once our final set of participants remain, we need to start merging the corresponding variables to our data set, variables such as Cognitive Status and the ones pertinent to each Risk Factor. This merge is possible thanks to the unique person identifier unhhidnp. Once everything is where it must be, the variables are both cleaned and treated to properly train our Random Forest Models.

This treatment is done by detecting and cleaning all risk-factor variables defined in the risk factors. It begins by collecting the expected variable names and matching them to the dataset using direct lookup, harmonized forms (converting wave-specific prefixes such as r3 to the unified form rW), substring searches, and household-level equivalents (hW). Once the relevant variables are identified, the function retains only these predictors along with identifiers, wave information, and the cognitive-status target. It then standardizes missing codes used in the MHAS files [27], converts ordinal text responses into numeric scales, and attempts to coerce remaining non-numeric variables into numerical form. Any columns still categorical are expanded through one-hot encoding to create binary indicator variables. Finally, the function builds a variable-to-group mapping that links each processed feature back to its corresponding risk-factor category, enabling grouped SHAP importance analysis.

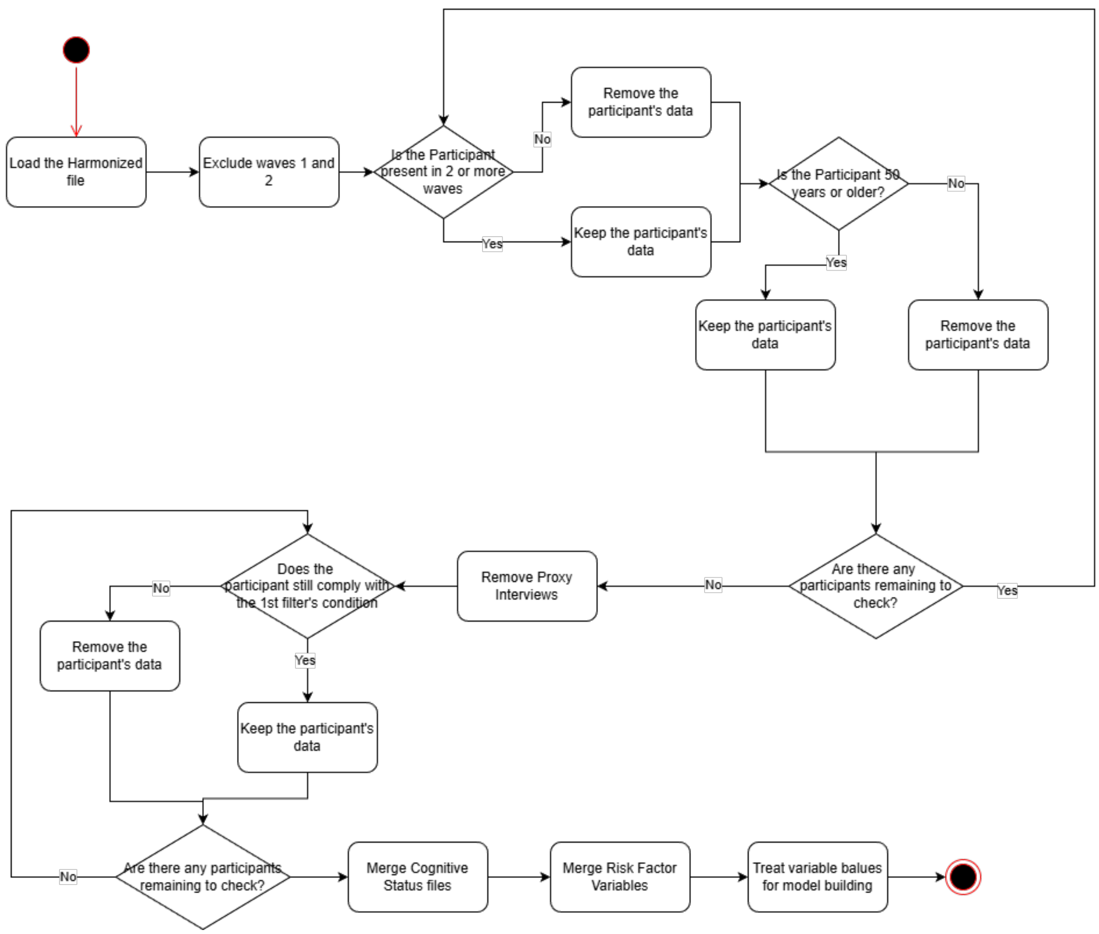

Figure 4.1: Initial Data Filtering

4.4.2 Model Training

Once the whole dataset has been treated (as described on Figure 4.1), we are ready to undergo training. The Random Forest is built, per wave, on 500 decision trees and an 80/20 spit for training. First we subset the data for the corresponding wave by having an x wave be all predictor variables for that wave and y wave the outcome or target variable. We then remove all the missing variable rows to ensure the models are trained on complete pairs and skip waves with the insufficient variation if there where any (on MHAS waves 3, 4, 5 and 6 there are no skips). After the training of the models is done, we need to compute the SHAP value across all samples and sort them. For each survey wave, the corresponding Random Forest model was reloaded and evaluated on the subset of participants from that wave. On each wave, the trained model and the relevant feature matrix and target variable were filtered from the data. Observations with missing outcomes were excluded, and waves with only one outcome class were skipped to avoid invalid metrics. Each model then generated predictions for its respective wave, and performance was assessed using precision, recall, F1-score, and accuracy. Additionally, multiclass discrimination ability was quantified by computing the one-vs-rest (OvR) ROC-AUC score, obtained through label binarization and probability-based predictions predict. When errors such as incompatible labels occurred, they were caught and reported without interrupting the loop.

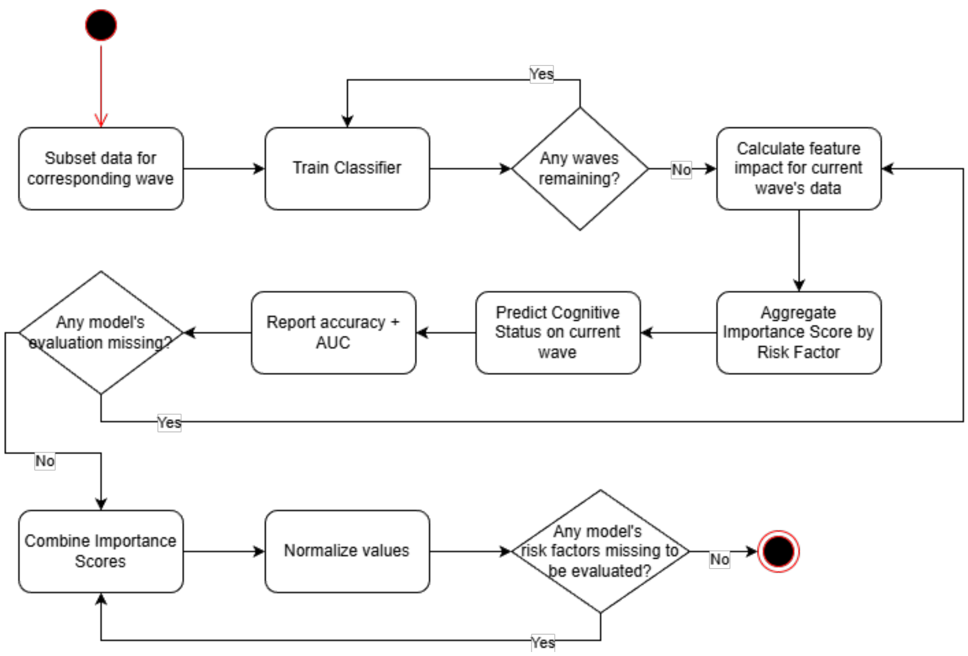

Figure 4.2: Model Training

4.4.3 Metrics and Diagnostics

Although not 'final' per se, our 3rd section of processing is the result extraction. On this section, we plot the evolution of our Risk Factors' importance, extract aging measurements per variable, and so on while also checking our data doesn't get duplicated, lost, etc. For starters we plot the distribution and percentage of Proxy Interviews, and Cognitive Status. Then, once the rest of the variables are merged, we chart the 'Missingness' of each variable alongside with the expected and found variables and values. Another metric that was useful to plot was Age. Form this variable we were able to extract values such as average age per wave, or number of Participants with 'x' age on 'y' wave. Thanks to these metrics we could plot answer distribution per variable, age distribution per answer and finally, a heat map of the correlation between Risk Factors. If the conditions for their respective codes have been executed, these plots and charts don't need to be extracted on any particular order.

Chapter 5

Results

5.1 Population Distribution

Our whole data set, after doing our filters (such as proxy interviews or 50 years of age) has this population distribution. Here, cognitive status normal appears as 0, normal with impairment as 1, CIND as 2 and Dementia as 3.

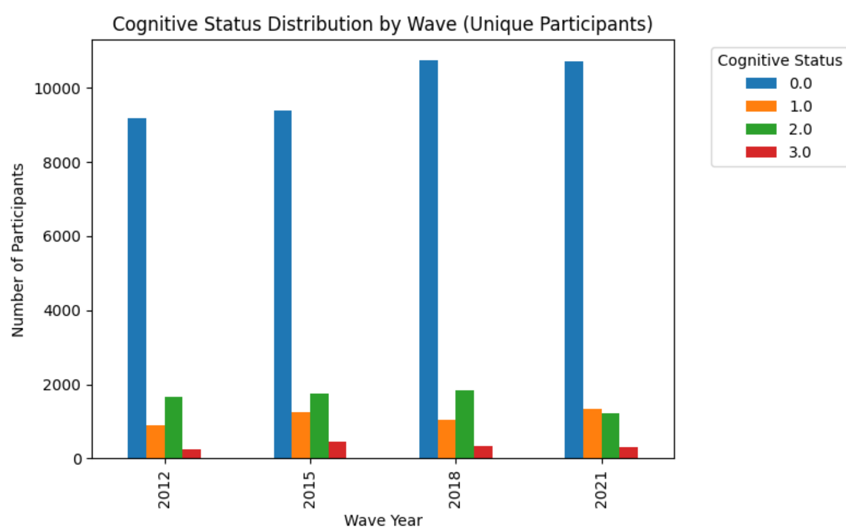

Figure 5.1: Cognitive Status Distribution

Cognitive Status	**Participants**	**Percent of Total**
0	17104	63.95%
1	3327	12.44%
2	5108	19.10%
3	1205	4.51%

Table 5.1: Overall Cognitive Status Distribution

Class 3, Dementia, is considerably smaller than the other classes, not even making it to 10%. This same behavior can be observed on each individual wave. Knowing this would prove useful for better model training, but that is discussed below.

5.2 Model Accuracy

During our research, several versions of the Random Forest models. Although we will remain with only one's results as final, it is important to report the other versions.

5.2.1 V1 - Post-filter Full Data Set

The first data set was trained on the whole Data Set after the filters were ran. Here, it was found that the data had a bias towards people with normal Cognitive status, as can be seen on Figure 5.1 and Table 5.1

Even with this population imbalance, the random forest model was trained, obtaining these results:

Class	**F1 Score**
0	94
1	73
2	76
3	61
Weighted Average	90

Table 5.2: Wave 3's model results

Class	**F1 Score**
0	94
1	81
2	79
3	77
Weighted Average	90

Table 5.3: Wave 4's model results

Class	**F1 Score**
0	95
1	77
2	76
3	68
Weighted Average	90

Table 5.4: Wave 5's model results

Class	**F1 Score**
0	95
1	80
2	67
3	70
Weighted Average	90

Table 5.5: Wave 6's model results

Here, we can observe that all the models have a weighted average f1 score of 90, and Class 0 (normal) has an f1 score of over 90. Although these may look like great percentages, in reality they show that these models risk over-fitting. This means that it has memorized it's training data too well, it has memorized the noise and irrelevant details instead of the underlying patterns. Because of this, we decided to train the next versions on a reduced Data Set.

5.2.2 V2 - Exactly Reduced Data Set

The next version reduced the other classes' population to exactly the same quantity as class 3, Dementia. The reductions were as follows:

Class	Original	Final
0	9823	261
1	905	261
2	1710	261
3	261	remains the same

Table 5.6: Wave 2012 Reduction (Target = 261)

Class	Original	Final
0	9856	462
1	1280	462
2	1814	462
3	462	remains the same

Table 5.7: Wave 2015 Reduction (Target = 462)

Class	Original	Final
0	11665	330
1	1070	330
2	1987	330
3	330	remains the same

Table 5.8: Wave 2018 Reduction (Target = 330)

Class	Original	Final
0	11080	320
1	1370	320
2	1249	320
3	320	remains the same

Table 5.9: Wave 2021 Reduction (Target = 320)

With this new population distribution, the model's results were as follows:

Class	F1 Score
0	89
1	89
2	90
3	90
Weighted Average	90

Table 5.10: Wave 3's model results

Class	F1 Score
0	87
1	89
2	88
3	89
Weighted Average	88

Table 5.11: Wave 4's model results

Class	F1 Score
0	88
1	88
2	88
3	88
Weighted Average	88

Table 5.12: Wave 5's model results

Class	F1 Score
0	88
1	89
2	87
3	87
Weighted Average	88

Table 5.13: Wave 6's model results

With the reduction, most of the models have a better weighted average. However, wave 3's model is still dangerously close to over-fitting. Thus, another type of reduction was made.

5.2.3 V3 - Double Amount Reduced Data Set

In order to better avoid over-fitting on all the models, instead of reducing each class' population to exactly class 3's, it was reduced to double the amount of class 3, which means the population distribution would look like this:

Class	Original	Final
0	9173	522
1	886	522
2	1660	522
3	261	remains the same

Table 5.14: Wave 2012 Reduction (Target = 522)

Class	Original	Final
0	9397	924
1	1266	924
2	1761	924
3	462	remains the same

Table 5.15: Wave 2015 Reduction (Target = 924)

Class	Original	Final
0	10754	660
1	1053	660
2	1851	660
3	330	remains the same

Table 5.16: Wave 2018 Reduction (Target = 660)

Class	Original	Final
0	10714	640
1	1350	640
2	1226	640
3	320	remains the same

Table 5.17: Wave 2021 Reduction (Target = 640)

With this second reduction, the results are as follows:

Class	F1 Score
0	0.88
1	0.89
2	0.89
3	0.89
Weighted Average	0.89

Table 5.18: Wave 2012 Random Forest Results

Class	F1 Score
0	0.90
1	0.90
2	0.89
3	0.89
Weighted Average	0.89

Table 5.19: Wave 2015 Random Forest Results

Class	F1 Score
0	0.90
1	0.90
2	0.90
3	0.89
Weighted Average	0.90

Table 5.20: Wave 2018 Random Forest Results

Class	F1 Score
0	0.89
1	0.90
2	0.88
3	0.88
Weighted Average	0.89

Table 5.21: Wave 2021 Random Forest Results

With this 2nd reduction, all models stray away from over-fitting, with none reaching 90% or more on their Weighted Average. These models were the ones used for obtaining the following graphs and results.

5.3 Predictor Importance

Thanks to the V3 models described on 5.2.3, we obtained the following in regards of the Risk Predictor importance and it's evolution throughout the years.

5.3.1 Top 5 Predictors

As the models are divided by wave, we can extract the top 5 Predictors and their importance this way too. They are as follows:

Group	Importance
Social Isolation	0.046198
Time Use	0.023501
Vision	0.021580
Obesity	0.019821
Hypertension	0.018092

Table 5.22: Wave 2012 – Top 5 SHAP Groups

Group	Importance
Time Use	0.050676
Social Isolation	0.042670
Hospitalization	0.025117
Hypertension	0.020247
Vision	0.019078

Table 5.23: Wave 2015 – Top 5 SHAP Groups

Group	Importance
Time Use	0.050090
Social Isolation	0.029894
Hypertension	0.028304
Education	0.018119
Falls	0.015637

Table 5.24: Wave 2018 – Top 5 SHAP Groups

Group	Importance
Time Use	0.050982
Social Isolation	0.039159
Education	0.020097
Falls	0.019615
Hypertension	0.019445

Table 5.25: Wave 2021 – Top 5 SHAP Groups

The whole assortment of SHAP Importance can be appreciated on this graph:

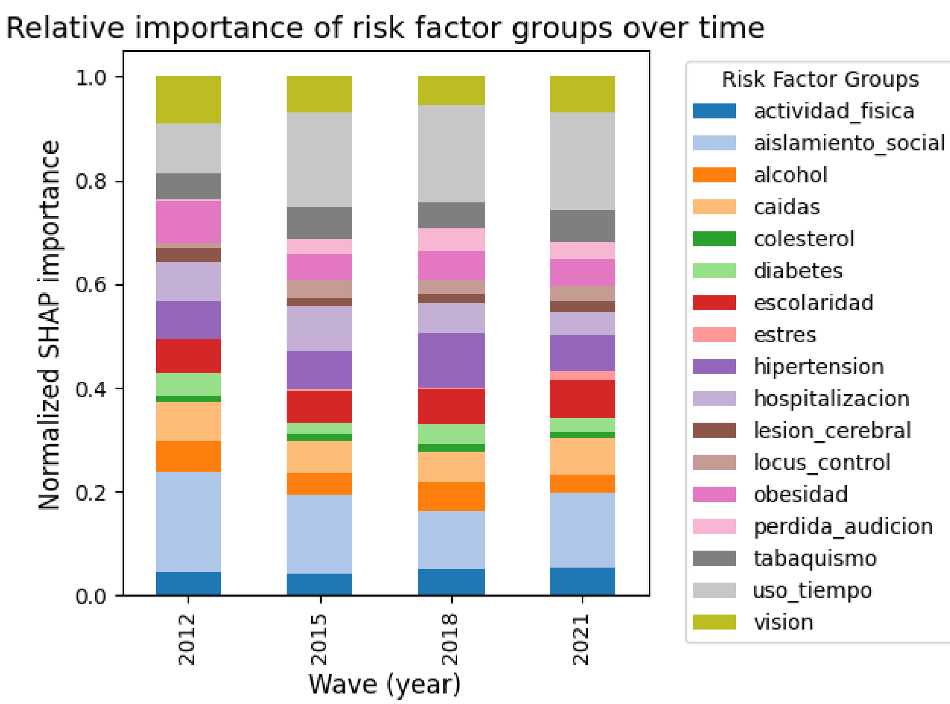

Figure 5.2: Relative Importance over Time

With these results we can see a pattern. Both Time Use and Social Isolation remain on the top 2 spots in the 4 waves. At most the most significant change we see is them switching places after wave 3. Other Risk Factors, although not as constant with their placing, do appear multiple times throughout the 4 waves. Hypertension appearing 3 times, and all Falls and Education and Vision appearing twice. There are only 2 outliers here that only appear once, these being Obesity and Hospitalization on wave 3 and wave 4 respectively.

5.3.2 Evolution Over Time

Even if we have identified the most important predictors per wave, we still need to analyze how this importance behaves through time. In order to better visualize it, we have the following Graph:

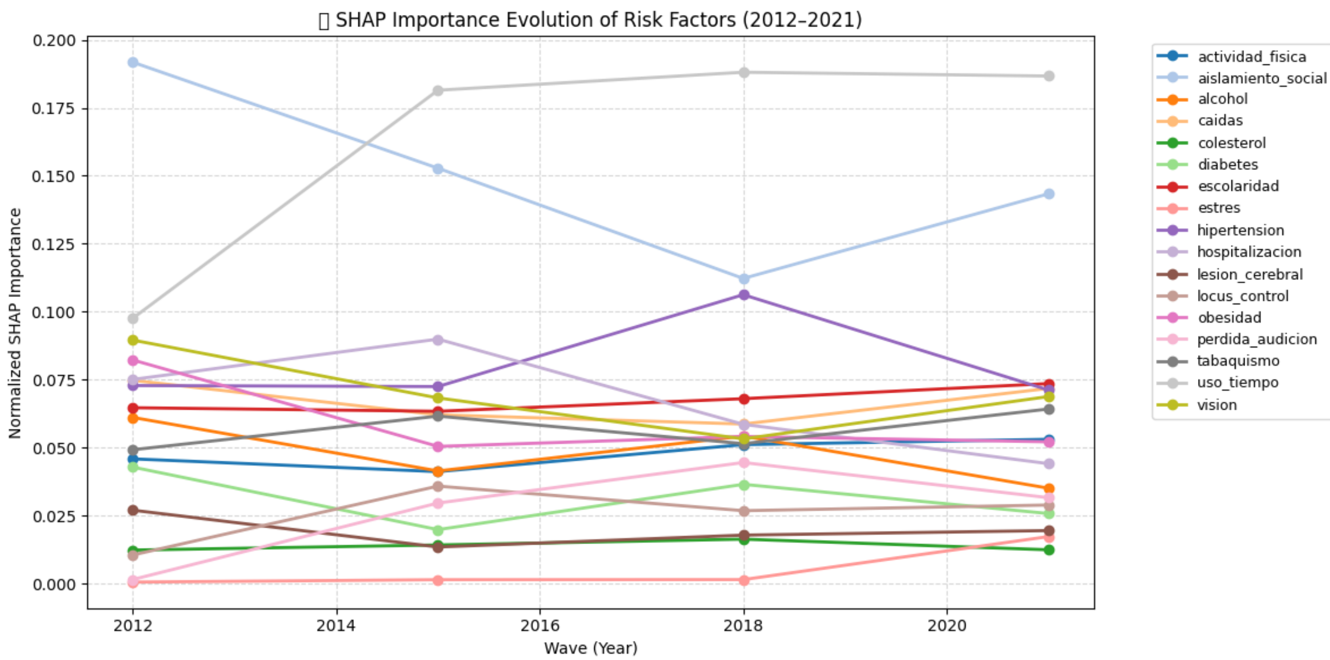

Figure 5.3: SHAP Importance Evolution

With Figure 5.3 we can better visualize how this importance behaves over time. For Example, we can see a rise in the importance of Social Isolation in between 2018 and 2021; this perhaps can be due to the global 2020 COVID-19 pandemic. Also due to the pandemic, Stress' importance rises on this same period of time too, as the variable that represents if the Respondent has been under stress due to COVID becomes available. Hypertension also is subjected to a sharp rise in Importance in the year 2018 only to even out again in 2021. Aside from those notable behaviors, most Risk Factors' importance remains constant throughout the 4 waves.

5.4 Validation

In order to reaffirm our ranking per wave, we have mixed the models in between the waves. 2012's model was used for wave 2015, 2015 on wave 2018, 2018 on 2021 and finally model 2021 in wave 2012. Here are both the top 5 predictors and the SHAP evolution throughout the waves based on this mixed analysis.

5.4.1 Top 5 Predictors

According to the mixed models' analysis, these are the top 5 predictors per wave:

Group	Importance
Time Use	0.053707
Social Isolation	0.041617
Hypertension	0.022078
Falls	0.021329
Vision	0.018549

Table 5.26: Cross-Wave SHAP: 2021 Model → 2012 Data

Group	Importance
Social Isolation	0.044962
Time Use	0.022989
Hospitalization	0.021174
Vision	0.020699
Obesity	0.018927

Table 5.27: Cross-Wave SHAP: 2012 Model → 2015 Data

Group	Importance
Time Use	0.050038
Social Isolation	0.042950
Hospitalization	0.025744
Hypertension	0.020803
Vision	0.019053

Table 5.28: Cross-Wave SHAP: 2015 Model → 2018 Data

Group	Importance
Time Use	0.050891
Social Isolation	0.029797
Hypertension	0.028519
Education	0.017689
Falls	0.015848

Table 5.29: Cross-Wave SHAP: 2018 Model → 2021 Data

5.4.2 Evolution Over Time

Finally, the evolution of our Risk Factors' SHAP Importance can be visualized here:

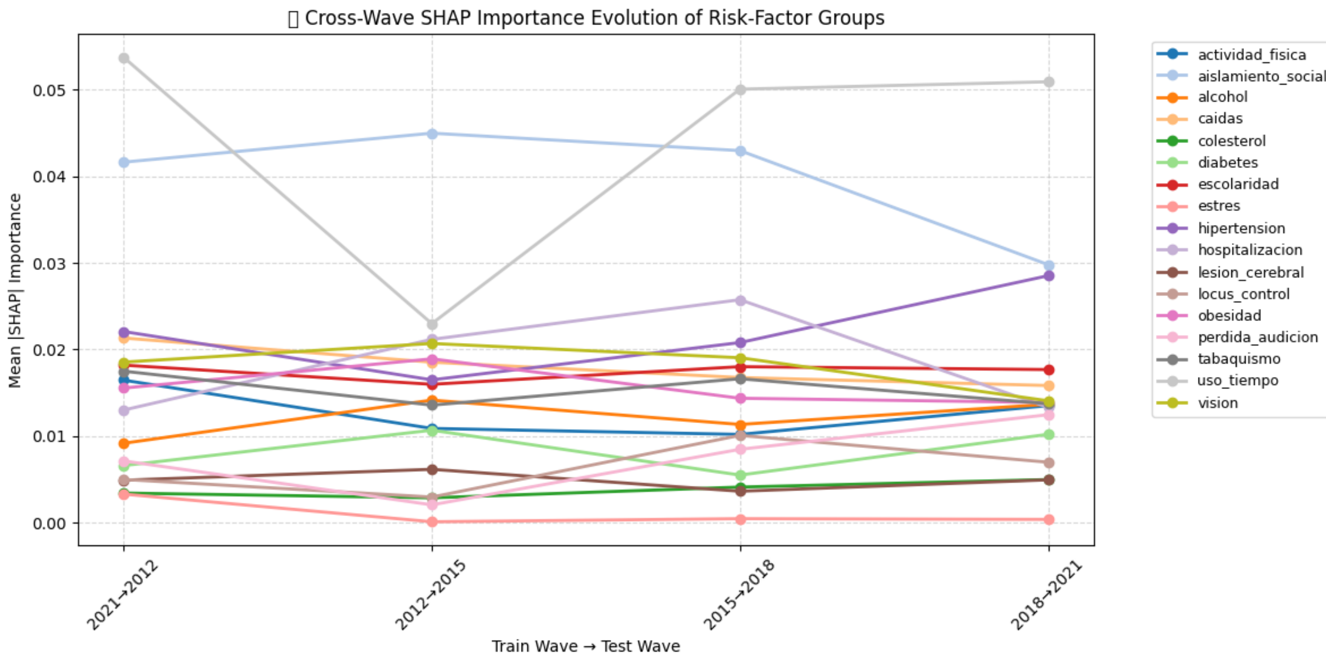

Figure 5.4: Mixed SHAP Importance Evolution

On Figure 5.4, the change in SHAP importance itself varies significantly from Figure 5.3, yet the Top 2 predictors remain the same. Not only but they maintain roughly the same behavior, with 2012's model predicting Social Isolation over Time Use. Not only that, but the rest of the top predictors remain mostly the same too, although varying in their placement.

Chapter 6

Conclusions

Thanks to our results and our validation, we could average out the Top 5 Predictors among these waves. The ranking is as follows:

1. Use of Time
2. Social Isolation
3. Hypertension
4. Education
5. Falls

With the Risk Factors Loss of Vision, Hospitalization and Obesity as possible places 6, 7 and 8. With this information in mind, if any other research team wishes to base something their own work on the MAHS' Database, now they can refer to our work. Another thing to conclude is that, maybe, once the 2024 data is released and available, we might even get to see Social Isolation either overtake Time Use once again or even out once more and remain 2nd place. We might even get to see Stress gain more importance, as that it started to raise it's importance after the global COVID-19 pandemic.

Chapter 7

Future Work

As mentioned in 3.2, Random Forest works better when taken a historical approach. Although this was stated, it wasn't done. The models we produced were trained exclusively on their own wave. A good way to better reaffirm the ranking we have obtained through our models would be to implement this historical approach. Instead of just plotting the importance through time based off of the results of each wave, a great next step would be to do this based on the whole dataset, without making the separation between waves. This wouldn't represent a change in methodology or implementation, but rather a new diagnostic/metric. As it was said, taking the historical approach afterwards would be more with the intention to reaffirm rather than affect our result.

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Appendix A

Variable Missingness

Below is TableA.1, which describes the found variables on each Risk Factor Group, how many answers were found and how many have missing data. Finally, it also shows the percentage missing of these variables

Table A.1: General Missingness Summary Across All Waves

Risk Factor Group	**Variable**	**Wave**	**Found**	**Missing**	**% Missing**
actividad_fisica	rWvigact	ALL	53378	27	0.05
aislamiento_social	hWrural_m	ALL	53405	0	0.00
aislamiento_social	rWcesd_m	ALL	53386	19	0.04
aislamiento_social	rWrfcntpmx_m	ALL	53302	103	0.19
alcohol	rWbinged	ALL	53341	64	0.12
alcohol	rWcage	ALL	5674	47731	89.38
alcohol	rWdrink	ALL	53388	17	0.03
alcohol	rWdrinkb	ALL	53341	64	0.12
alcohol	rWdrinkd	ALL	53160	245	0.46
alcohol	rWdrinkn	ALL	53282	123	0.23
caidas	rWfall	ALL	53381	24	0.04
caidas	rWfallinj	ALL	39138	14267	26.71
caidas	rWfallnum	ALL	53251	154	0.29
colesterol	rWcholst	ALL	53329	76	0.14
contaminacion_aire	hWrural	ALL	53405	0	0.00
contaminacion_aire	hWrural_m	ALL	53405	0	0.00
diabetes	rWdiabe	ALL	53332	73	0.14
diabetes	rWrxdiab	ALL	53320	85	0.16
escolaridad	raedyrs	ALL	52824	581	1.09
escolaridad	raliterate	ALL	52776	629	1.18
escolaridad	ranumerate	ALL	52757	648	1.21
estres	hWchdeathe	ALL	53405	0	0.00

Continued on next page

Table A.1 – continued from previous page

Risk Factor Group	Variable	Wave	Found	Missing	% Missing
estres	rWcconc_m	ALL	13744	39661	74.26
estres	rWunemp	ALL	21694	31711	59.38
hipertension	rWhibpe	ALL	53351	54	0.10
hipertension	rWrxhibp	ALL	53336	69	0.13
hospitalizacion	rWhosp1y	ALL	53332	73	0.14
hospitalizacion	rWhspnit1y	ALL	53332	73	0.14
lesion_cerebral	rWrecstrok	ALL	1889	51516	96.46
lesion_cerebral	rWrxstrok	ALL	53365	40	0.07
lesion_cerebral	rWstroke	ALL	53371	34	0.06
locus_control	rWlsatsc3m	ALL	41089	12316	23.06
locus_control	rWsatlifez	ALL	53113	292	0.55
obesidad	rWbmi	ALL	47679	5726	10.72
obesidad	rWobese	ALL	47679	5726	10.72
perdida_audicion	rWhearaid	ALL	53398	7	0.01
perdida_audicion	rWhearing	ALL	41146	12259	22.95
tabaquismo	rWquitsmok	ALL	19272	34133	63.91
tabaquismo	rWsmokef	ALL	53368	37	0.07
tabaquismo	rWsmokefm	ALL	47172	6233	11.67
tabaquismo	rWsmoken	ALL	53378	27	0.05
tabaquismo	rWsmokev	ALL	53381	24	0.04
tabaquismo	rWstrtsmok	ALL	25520	27885	52.21
uso_tiempo	rWslfemp	ALL	20556	32849	61.51
uso_tiempo	rWsocact_m	ALL	53380	25	0.05
uso_tiempo	rWwork	ALL	53291	114	0.21
vision	rWglasses	ALL	53391	14	0.03
vision	rWsight	ALL	53047	358	0.67

Appendix B

Age Distributions

This section covers the general age distribution per wave on Figures B.1, B.2, B.3 and B.4. They are displayed on bar graphs to show the quantity of each age identified per wave.

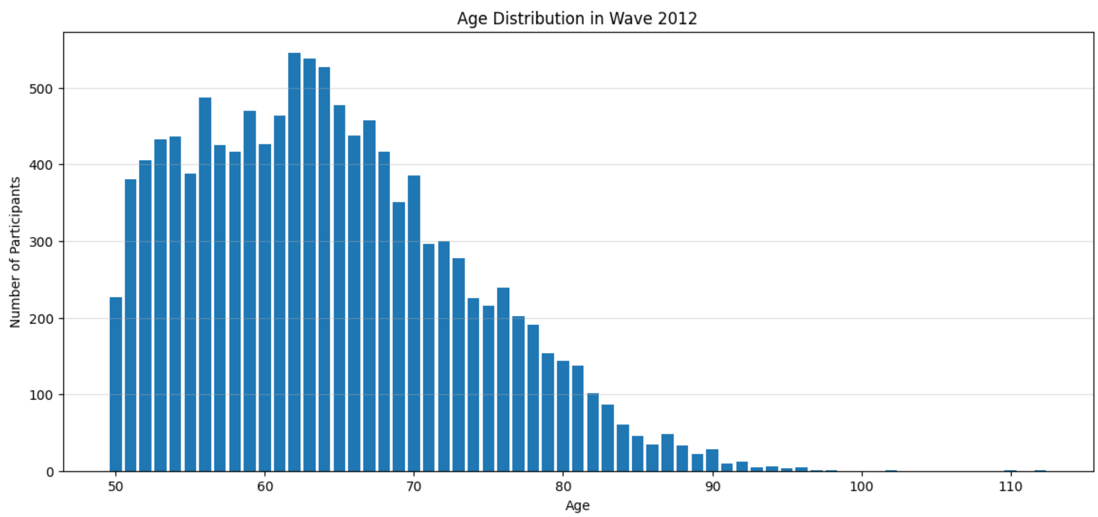

Figure B.1: Age Distribution on Wave 3

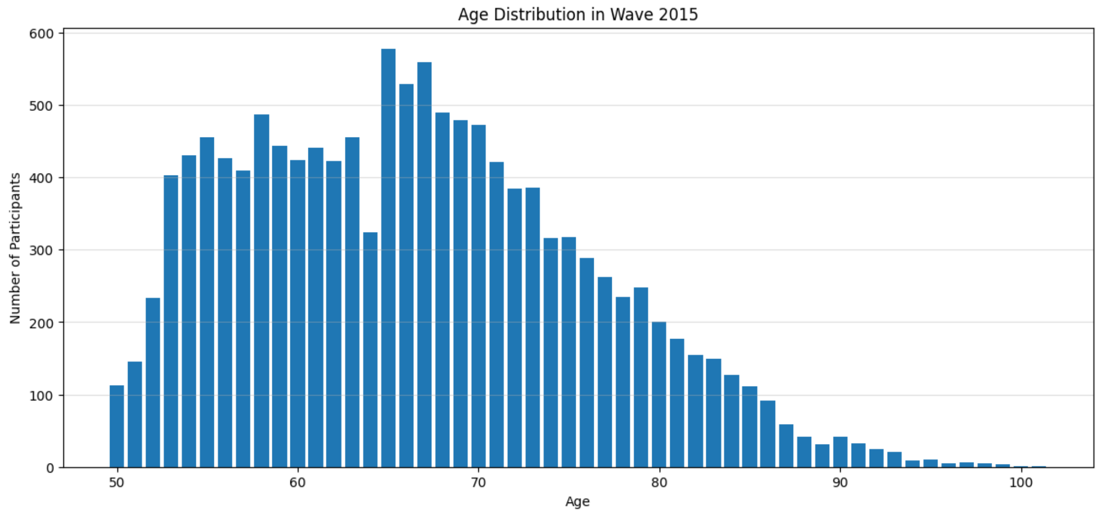

Figure B.2: Age Distribution on Wave 4

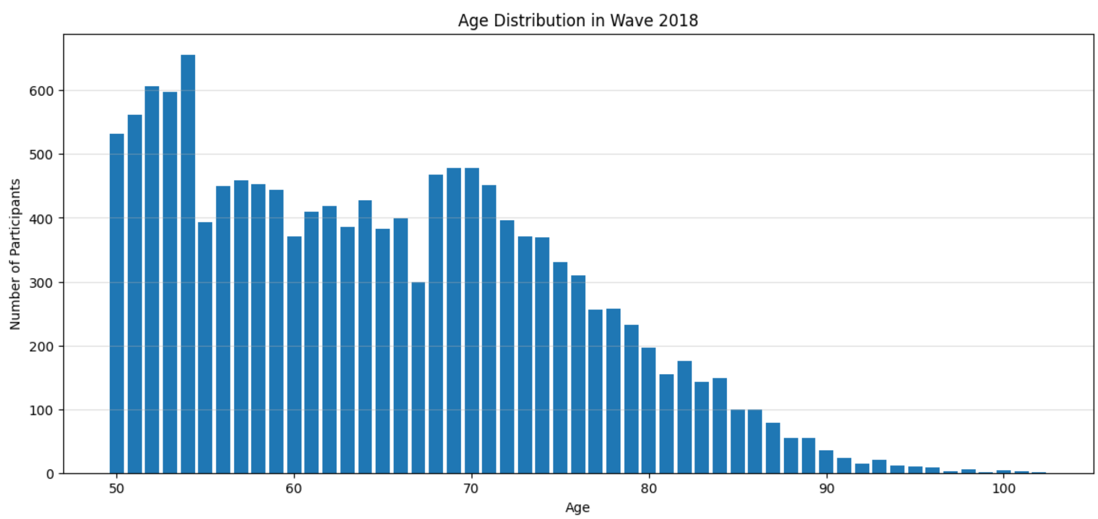

Figure B.3: Age Distribution on Wave 5

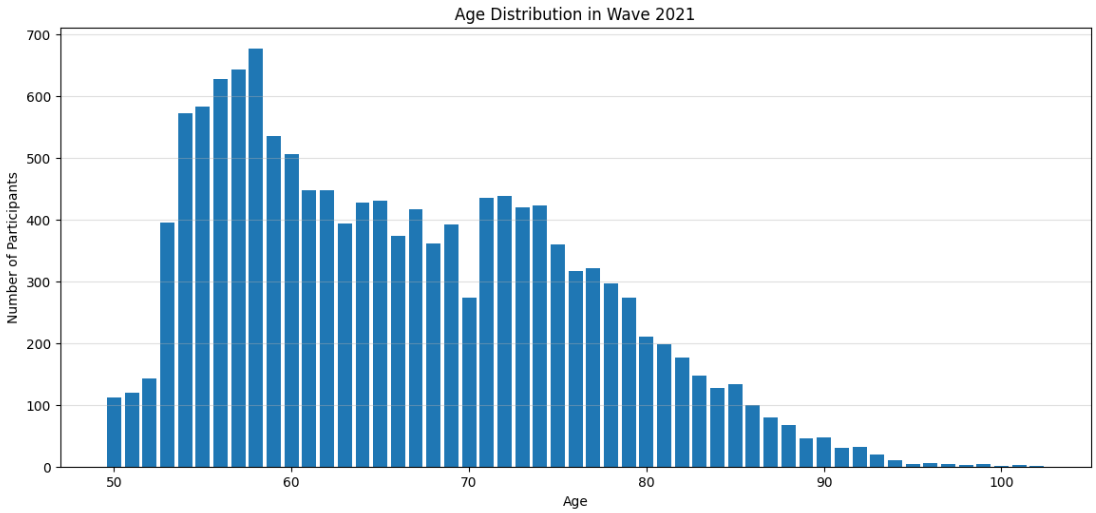

Figure B.4: Age Distribution on Wave 6

Appendix C

Persistent and Emergent Predictors

As part of the metrics extracted during the process of training, we have also extracted Tables C.1 and C.2 as a way to represent the importance gain for the Emergent predictors and the Average Importance of the Persistent ones.

Table C.1: Persistent Risk Factor Importance

Risk Factor	**Importance**
Uso del Tiempo	0.176170
Aislamiento Social	0.147949
Hipertensión	0.085306
Caídas	0.079733
Escolaridad	0.076428
Visión	0.066791
Tabaquismo	0.058693
Hospitalización	0.054666
Obesidad	0.052951
Alcohol	0.047711

Table C.2: Risk Factor Importance Gain

Risk Factor	**Importance Gain**
Uso del Tiempo	0.048354
Actividad Física	0.042953
Pérdida de Audición	0.026819
Estrés	0.021903
Hipertensión	0.013858
Colesterol	0.010870
Hospitalización	0.005404
Tabaquismo	0.001747
Lesión Cerebral	0.000566
Obesidad	0.000471

Appendix D

Variable Correlations

Another one of the metrics we have extracted were these heatmaps that describe the correlation between variables, if any.

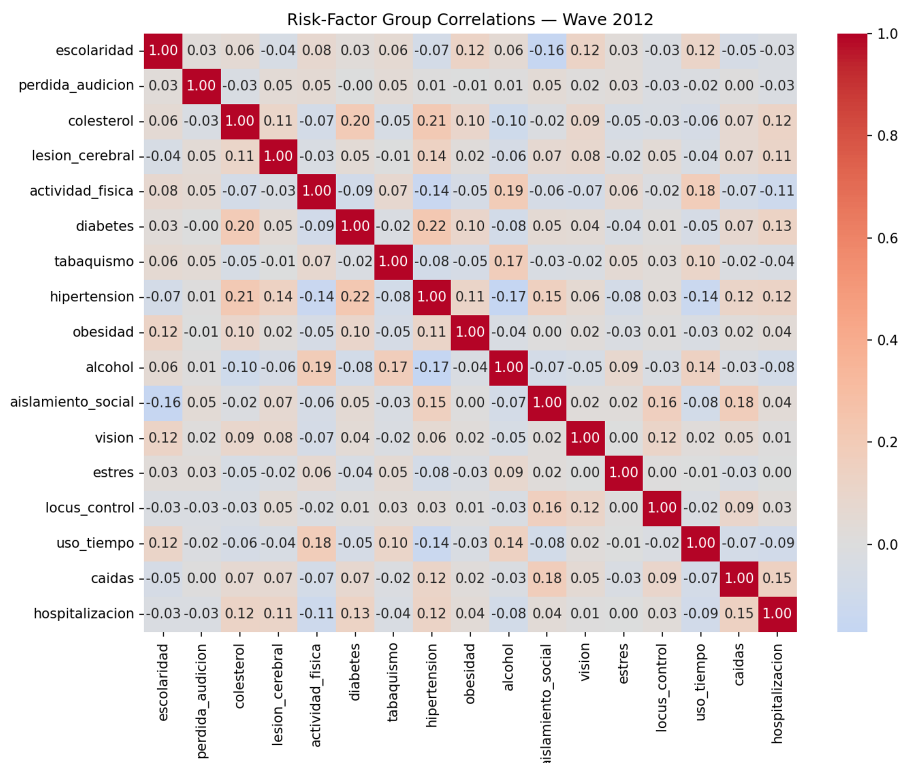

Figure D.1: Variable Correlation on Wave 3

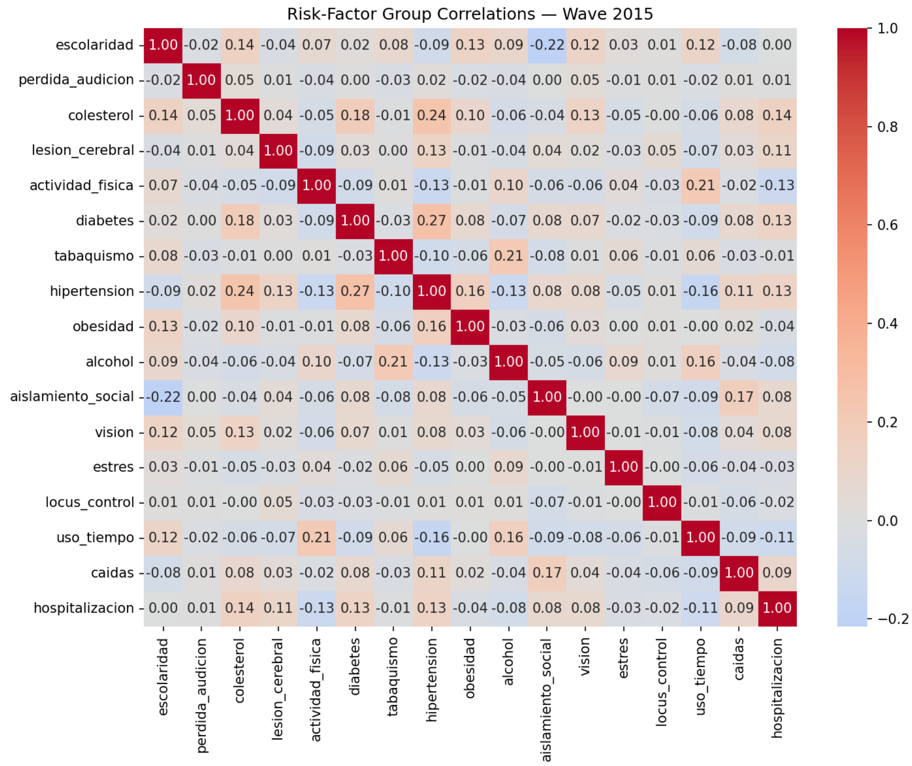

Figure D.2: Variable Correlation on Wave 4

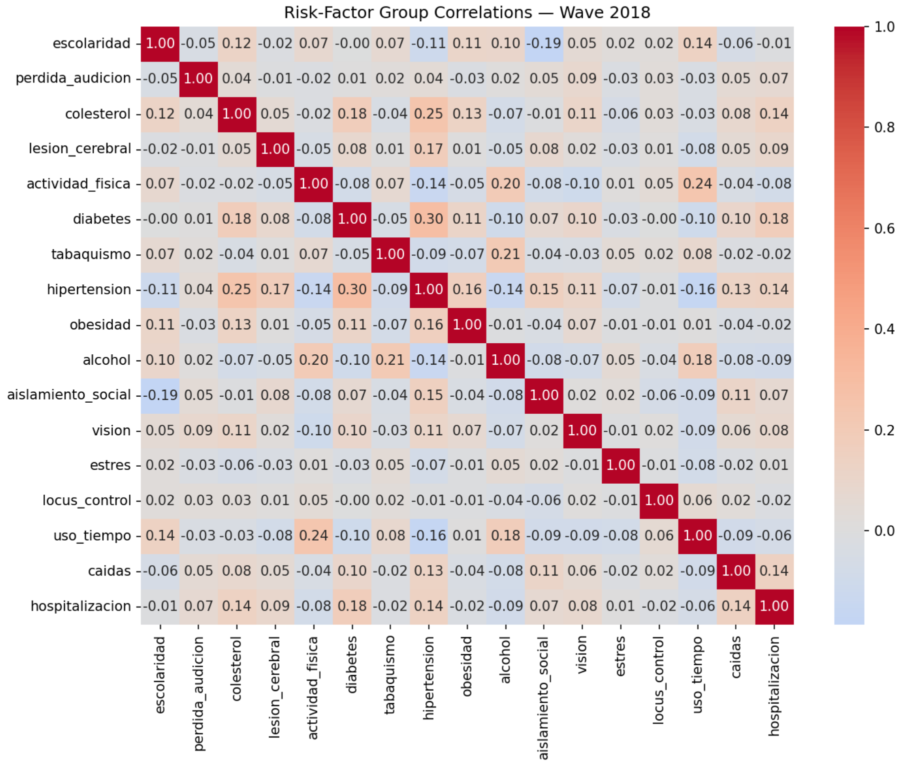

Figure D.3: Variable Correlation on Wave 5

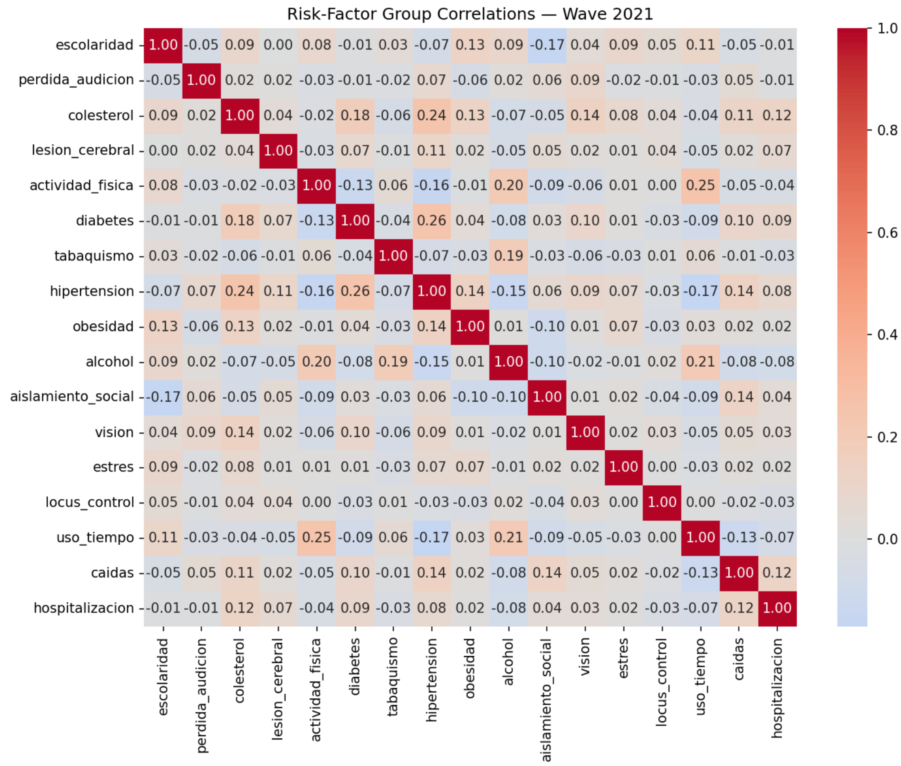

Figure D.4: Variable Correlation on Wave 6

Appendix E

Files

The files corresponding to the RF models trained per wave, the results their analysis obtained and the final dataset can be found on the following link:

https://drive.google.com/drive/folders/1FLV_955mLWJp0_GOWdgj_ummXMHunmlL?usp=sharing